

MAT1002: Calculus II L01

Quiz 3 — 2025-2026, Term 2

Tuesday, 7 Apr 2026

1. (a) (5 points) Show that the following limit does not exist,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}.$$

- (b) (5 points) Does

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2},$$

exist? If so, prove it; if not, justify your answer.

Solution:

- (a) Along the curve $x = ky^2$, $(x, y) \rightarrow 0$ as $y \rightarrow 0$, so

$$\lim_{(x,y) \rightarrow (0,0) \text{ along } x=ky^2} \frac{xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0} \frac{ky^4}{(k^2 + 1)y^4} = \frac{k}{k^2 + 1} = \begin{cases} \frac{1}{2}, & k = 1, \\ -\frac{1}{2}, & k = -1. \end{cases}$$

Hence, the limit does not exist.

- (b) For arbitrary $\epsilon > 0$, we let $\delta = \epsilon/3$, then for all (x, y) with $0 < \sqrt{x^2 + y^2} < \delta$, we have

$$\left| \frac{3x^2y}{x^2 + y^2} \right| < 3 \left| \frac{x^2y}{x^2} \right| < 3|y| \leq 3\sqrt{x^2 + y^2} < 3\delta = \epsilon,$$

so

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0.$$

Students may use other approaches to prove it.

2. (a) (4 points) Suppose that $z = e^x \sin y$, $x = st^2$ and $y = s^2t$. Find $\partial z / \partial s$ and $\partial z / \partial t$.
(b) (3 points) Find the slope of the tangent line to the curve $x^3 + y^3 = 6xy$ at the point $(3, 3)$.
(c) (3 points) Find the directional derivative of $f(x, y) := xe^y + \cos(xy)$ at the point $(2, 0)$ in the direction of $\vec{v} = \langle 3, -4 \rangle$.

Solution:

- (a)

$$\frac{\partial z}{\partial s} = t^2 \sin(s^2t) e^{st^2} + 2st \cos(s^2t) e^{st^2}.$$

$$\frac{\partial z}{\partial t} = 2st \sin(s^2 t) e^{st^2} + s^2 \cos(s^2 t) e^{st^2}.$$

(b) Let $F(x, y) = x^3 + y^3 - 6xy$, so $F_x = 3x^2 - 6y$ and $F_y = 3y^2 - 6x$. Then

$$\left. \frac{dy}{dx} \right|_{(x,y)=(3,5)} = -\frac{F_x(3, 3)}{F_y(3, 3)} = -1.$$

(c) The unit vector along $\vec{v}$ is $\hat{v} = \langle 3/5, -4/5 \rangle$. The directional derivative of f at $(2, 0)$ in the direction of $\hat{v}$ is

$$\begin{aligned} D_{\vec{v}} f(2, 0) &= \lim_{h \rightarrow 0} \frac{f(2 + 3h/5, -4h/5) - f(2, 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(2 + 3h/5)e^{-4h/5} + \cos[(2 + 3h/5)(-4h/5)] - (2 + 1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2(e^{-4h/5} - 1)}{h} + \lim_{h \rightarrow 0} \frac{3}{5} e^{-4h/5} + \lim_{h \rightarrow 0} \frac{\cos(-8h/5 - 12h^2/5) - 1}{h} \\ &= 2 \left. \frac{d[e^{-4h/5}]}{dh} \right|_{h \rightarrow 0} + \frac{3}{5} + 2 \left. \frac{d[\cos(-8h/5 - 12h^2/5)]}{dh} \right|_{h \rightarrow 0} \\ &= -\frac{8}{5} + \frac{3}{5} \\ &= -1. \end{aligned}$$

Alternatively, we can directly calculate

$$D_{\hat{v}} f(2, 0) = \nabla f(2, 0) \cdot \hat{v} = \langle 1, 2 \rangle \cdot \left\langle \frac{3}{5}, -\frac{4}{5} \right\rangle = -1.$$

3. (a) (6 points) For $f(x, y, z) = x^2 - xy + 3 \sin z$, consider the standard linear approximation at $P_0(2, 1, 0)$ on

$$R = [1.99, 2.01] \times [0.98, 1.02] \times [-0.01, 0.01].$$

Find the linear approximation from P_0 in R and estimate the error.

- (b) (4 points) Find all critical points of the function

$$f(x, y) := x^4 + y^4 - 4xy + 1,$$

defined on $\mathbb{R}^2$.

Solution:

- (a) At P_0 , we have

$$f(P_0) = 2, f_x(P_0) = 3, f_y(P_0) = -2, f_z(P_0) = 3.$$

The linear approximation is

$$L(x, y, z) = 2 + 3(x - 2) - 2(y - 1) + 3z = 3x - 2y + 3z - 2.$$

Since $f_{xx} = 2$, $f_{yy} = 0$, $f_{zz} = -3 \sin z$, $f_{xy} = -1$, $f_{xz} = 0$, $f_{yz=0} = 0$, all their absolute values are ≤ 2 . Hence, for all (x, y, z) in R ,

$$|E(x, y, z)| \leq \frac{1}{2} \cdot 2(|x - 2| + |y - 1| + |z - 0|)^2 \leq (0.01 + 0.02 + 0.01)^2 = 0.0016.$$

(b) Solving

$$f_x = 4x^3 - 4y = 0, f_y = 4y^3 - 4x = 0,$$

we have three solutions (critical points): $(x, y) = (1, 1)$ or $(x, y) = (-1, -1)$ or $(x, y) = (0, 0)$.